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Person-Centered Methods in STEM Education: The Gap Between *Methodological Promise* and *Equity Practice*

Based on: "What Does the 'Person-Centered Quantitative Analysis' Mean in Practice? A Literature Review in STEM Education Research"

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01 INTRODUCTION

Despite decades of reform, higher education remains stratified by race, class, and gender [1]. In quantitative research, variable-centered methods often emphasize broad generalizations, risk treating social categories as fixed, and may obscure meaningful subgroup differences.

Person-centered quantitative approaches, which identify latent subgroups rather than averaging across populations, have been advocated as equity-oriented tools that can reveal within-group heterogeneity and move beyond deficit-oriented demographic binning.

But has this promise been realized? This study systematically reviews how person-centered approaches have been applied in STEM education research and finds a gap between methodological promise and actual practice.

02 WHAT IS PERSON-CENTERED ANALYSIS? [2]

VARIABLE-CENTERED

Examines relationships among variables (regression, SEM). Assumes population homogeneity. Powerful for average trends, but may obscure subgroup differences.

PERSON-CENTERED

Seeks to identify subgroups of individuals who share similar patterns across a set of observed variables (LPA, LCA, cluster analysis, mixture models).

PERSON-SPECIFIC

Focuses on within-individual variability over time (dynamic factor analysis, state-space modeling). A third paradigm.

BY THE NUMBERS

156 INITIAL RESULTS

91 AFTER SCREENING

19 FURTHER SCREENING

13 STUDIES INCLUDED

03 METHOD

Systematic literature review following PRISMA guidelines [3], drawing on Scopus. An education-focused keyword search yielded 156 articles; screening for date, scholarly & English sources, quantitative analysis, and STEM-education focus retained 13 empirical studies.

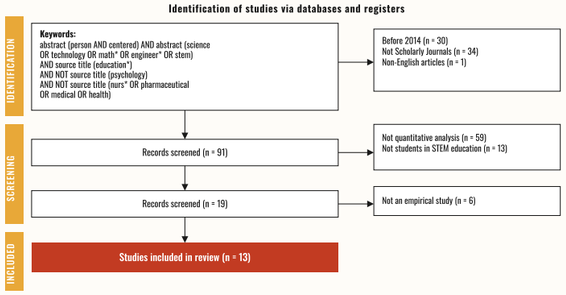


FIG 1. PRISMA flow diagram for the systematic literature review.

04 LIMITATIONS & FUTURE WORK

Relying solely on Scopus with narrow title exclusions may have limited scope. Future work will add databases, broaden search strategies, and reach interdisciplinary contexts. New methods (topological data analysis, ML-based clustering) may reshape how PCA is operationalized.

05 KEY FINDINGS

Studies mainly identified learner profiles around motivation, emotion, cognition, and identity, linking them to engagement, achievement, and career outcomes.

Students with strong self-regulation outperformed peers with high test anxiety; distinct motivational profiles were associated with science career expectations.

Contextual factors like teacher support and perceived fairness shifted profiles over time; gender, interventions, and engineering identity affected classification stability.

SAMPLE CHARACTERISTICS

Sample sizes vary substantially across studies, from 77 Chilean ninth-grade students to 6,778 primary-level students from Norway, Portugal, and Serbia. Pre-college studies draw on international samples (Sweden, Chile, multiple PISA countries), while college-level studies are predominantly based in U.S. institutions, with one exception from the Netherlands.

ANALYTICAL APPROACHES

Most studies used Latent Profile Analysis or Latent Class Analysis to identify learner profiles. LPA was commonly applied to group students based on motivational, epistemic, ICT-related, and emotional factors. LCA was used to examine science identity classifications.

Person-centered approaches hold equity potential, but the reviewed studies suggest this potential remains largely untapped. Most studies stop at describing learner profiles without explicitly connecting their findings to equity concerns. QuantCrit provides a lens to understand why [4].

06 QUANTCRIT DIAGNOSIS: WHY THE GAP EXISTS

1. CENTRALITY OF RACISM	The method itself does not ask about racism. It will find groups whether or not the researcher has thought about how race shapes the data.
2. NUMBERS ARE NOT NEUTRAL	The choice of what to measure and how to measure it reflects assumptions. If those choices are not examined critically, the resulting profiles inherit built-in biases.
3. CATEGORIES ARE NOT NATURAL	PCA can move past simple labels like "White vs. non-White," but the new groups it finds can be mistaken for fixed types rather than patterns shaped by context.
4. DATA CANNOT SPEAK FOR ITSELF	A group labeled "High Performance, Low Belonging" could mean students lack belonging, or that institutions push them out. The label depends on who interprets it.
5. NUMBERS FOR SOCIAL JUSTICE	Most studies stop at describing groups. Few are designed from the start to ask equity questions or inform change.

However, QuantCrit itself emerged from U.S. racial contexts. Global engineering education inequities operate along axes beyond race: class, caste, colonialism, hukou. Critical frameworks must be adapted, not mechanically transferred.

07 STAKEHOLDERS & NEEDS

STUDENTS

Students need their diverse experiences to be captured accurately, not reduced to demographic bins or group averages. PCA can identify subgroups whose needs are otherwise overlooked in traditional analyses.

RESEARCHERS

Researchers need clearer methodological guidance for choosing person-centered approaches and accessible models for integrating equity frameworks into quantitative design. PCA can serve equity goals when guided by intentional research questions.

ACADEMIC INSTITUTIONS

Academic institutions need review standards that recognize equity engagement, not only statistical rigor, and platforms for cross-disciplinary dialogue. PCA can bridge quantitative and critical traditions when supported institutionally.

POLICYMAKERS

Policymakers need structural explanations for why student subgroups form, not just descriptions of what groups exist. PCA can reveal which modifiable conditions shape outcomes, enabling targeted interventions.

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